Lab Sheet 4 - Solution

SQL Operators

Objective: To practice arithmetic, logical, comparison, special and set operators in MySQL.

Note: Every query was executed and the output shown is the real result, on a MySQL-compatible server (MariaDB 10.11). INTERSECT and EXCEPT need MySQL 8.0.31 or later; on older MySQL use `IN` / `NOT IN` with a subquery instead (for example `WHERE City IN (SELECT City FROM Customer)`).

Setup (run first)

Create the database

SQL

```
DROP DATABASE IF EXISTS CollegeDB;
CREATE DATABASE CollegeDB;
```

Output

```
Query OK, 2 rows affected

Query OK, 1 row affected
```

Create tables and load sample data

SQL

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name       VARCHAR(50),
    Age        INT,
    Course     VARCHAR(30),
    City       VARCHAR(30)
);
CREATE TABLE Employee (
    Emp_ID     INT PRIMARY KEY,
    Emp_Name   VARCHAR(50),
    Department VARCHAR(30),
    Salary     DECIMAL(10,2),
    Bonus      DECIMAL(10,2)          -- NULL means no bonus
);
CREATE TABLE Product (
    Product_ID   INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Price        DECIMAL(10,2),
    Quantity     INT
);
CREATE TABLE Customer (
    Customer_ID   INT PRIMARY KEY,
    Customer_Name VARCHAR(50),
    City          VARCHAR(30)
);
INSERT INTO Student VALUES
    (101, 'Rahul Sharma', 20, 'B.Tech', 'Meerut'),
    (102, 'Priya Singh',  21, 'BCA',    'Delhi'),
    (103, 'Amit Verma',   22, 'MCA',    'Roorkee'),
    (104, 'Neha Gupta',   19, 'B.Tech', 'Delhi'),
    (105, 'Karan Mehta',  23, 'B.Tech', 'Meerut'),
    (106, 'Riya Joshi',   20, 'BCA',    'Noida'),
    (107, 'Rohit Jain',   22, 'BCA',    'Delhi');
INSERT INTO Employee VALUES
    (1, 'Arun Kapoor',  'IT',      55000, 5000),
    (2, 'Meena Iyer',   'HR',      42000, NULL),
    (3, 'Suresh Patel', 'Finance', 48000, 3000),
    (4, 'Divya Menon',  'IT',      61000, NULL),
    (5, 'Ravi Das',     'Sales',   30000, 2000),
    (6, 'Rekha Nair',   'HR',      45000, 4000);
INSERT INTO Product VALUES
    (1, 'Laptop',   55000, 10),
    (2, 'Mouse',      500, 50),
    (3, 'Keyboard',  1200, 25),
    (4, 'Monitor',   9000, 15),
    (5, 'Webcam',    2500,  8);
INSERT INTO Customer VALUES
    (1, 'Anil Khanna',  'Delhi'),
    (2, 'Sunita Rao',   'Mumbai'),
    (3, 'Manoj Tiwari', 'Meerut'),
    (4, 'Kavita Shah',  'Pune');
SELECT * FROM Student;
SELECT * FROM Employee;
SELECT * FROM Product;
SELECT * FROM Customer;
```

Output

```
Query OK, 0 rows affected

Query OK, 0 rows affected

Query OK, 0 rows affected

Query OK, 0 rows affected

Query OK, 7 rows affected
Records: 7  Duplicates: 0  Warnings: 0

Query OK, 6 rows affected
Records: 6  Duplicates: 0  Warnings: 0

Query OK, 5 rows affected
Records: 5  Duplicates: 0  Warnings: 0

Query OK, 4 rows affected
Records: 4  Duplicates: 0  Warnings: 0

+------------+--------------+------+--------+---------+
| Student_ID | Name         | Age  | Course | City    |
+------------+--------------+------+--------+---------+
|        101 | Rahul Sharma |   20 | B.Tech | Meerut  |
|        102 | Priya Singh  |   21 | BCA    | Delhi   |
|        103 | Amit Verma   |   22 | MCA    | Roorkee |
|        104 | Neha Gupta   |   19 | B.Tech | Delhi   |
|        105 | Karan Mehta  |   23 | B.Tech | Meerut  |
|        106 | Riya Joshi   |   20 | BCA    | Noida   |
|        107 | Rohit Jain   |   22 | BCA    | Delhi   |
+------------+--------------+------+--------+---------+
7 rows in set

+--------+--------------+------------+----------+---------+
| Emp_ID | Emp_Name     | Department | Salary   | Bonus   |
+--------+--------------+------------+----------+---------+
|      1 | Arun Kapoor  | IT         | 55000.00 | 5000.00 |
|      2 | Meena Iyer   | HR         | 42000.00 |    NULL |
|      3 | Suresh Patel | Finance    | 48000.00 | 3000.00 |
|      4 | Divya Menon  | IT         | 61000.00 |    NULL |
|      5 | Ravi Das     | Sales      | 30000.00 | 2000.00 |
|      6 | Rekha Nair   | HR         | 45000.00 | 4000.00 |
+--------+--------------+------------+----------+---------+
6 rows in set

+------------+--------------+----------+----------+
| Product_ID | Product_Name | Price    | Quantity |
+------------+--------------+----------+----------+
|          1 | Laptop       | 55000.00 |       10 |
|          2 | Mouse        |   500.00 |       50 |
|          3 | Keyboard     |  1200.00 |       25 |
|          4 | Monitor      |  9000.00 |       15 |
|          5 | Webcam       |  2500.00 |        8 |
+------------+--------------+----------+----------+
5 rows in set

+-------------+---------------+--------+
| Customer_ID | Customer_Name | City   |
+-------------+---------------+--------+
|           1 | Anil Khanna   | Delhi  |
|           2 | Sunita Rao    | Mumbai |
|           3 | Manoj Tiwari  | Meerut |
|           4 | Kavita Shah   | Pune   |
+-------------+---------------+--------+
4 rows in set
```

A. Arithmetic Operators

Q1. Perform addition of two numbers using +.

SQL

```
SELECT 50 + 30 AS Addition;
```

Output

```
+----------+
| Addition |
+----------+
|       80 |
+----------+
1 row in set
```

Q2. Perform subtraction using -.

SQL

```
SELECT 100 - 45 AS Subtraction;
```

Output

```
+-------------+
| Subtraction |
+-------------+
|          55 |
+-------------+
1 row in set
```

*Q3. Perform multiplication using *.*

SQL

```
SELECT 12 * 8 AS Multiplication;
```

Output

```
+----------------+
| Multiplication |
+----------------+
|             96 |
+----------------+
1 row in set
```

Q4. Perform division using /.

SQL

```
SELECT 100 / 8 AS Division, 100 DIV 8 AS Integer_Division, 100 % 8 AS Remainder;
```

Output

```
+----------+------------------+-----------+
| Division | Integer_Division | Remainder |
+----------+------------------+-----------+
|  12.5000 |               12 |         4 |
+----------+------------------+-----------+
1 row in set
```

Q5. Calculate the total amount as Price x Quantity.

SQL

```
SELECT Product_Name, Price, Quantity, Price * Quantity AS Total_Amount FROM Product;
```

Output

```
+--------------+----------+----------+--------------+
| Product_Name | Price    | Quantity | Total_Amount |
+--------------+----------+----------+--------------+
| Laptop       | 55000.00 |       10 |    550000.00 |
| Mouse        |   500.00 |       50 |     25000.00 |
| Keyboard     |  1200.00 |       25 |     30000.00 |
| Monitor      |  9000.00 |       15 |    135000.00 |
| Webcam       |  2500.00 |        8 |     20000.00 |
+--------------+----------+----------+--------------+
5 rows in set
```

B. Logical Operators

Q6. Demonstrate AND with two conditions.

SQL

```
SELECT * FROM Student WHERE Age > 20 AND City = 'Delhi';
```

Output

```
+------------+-------------+------+--------+-------+
| Student_ID | Name        | Age  | Course | City  |
+------------+-------------+------+--------+-------+
|        102 | Priya Singh |   21 | BCA    | Delhi |
|        107 | Rohit Jain  |   22 | BCA    | Delhi |
+------------+-------------+------+--------+-------+
2 rows in set
```

Q7. Demonstrate OR with two conditions.

SQL

```
SELECT * FROM Student WHERE City = 'Noida' OR Age = 23;
```

Output

```
+------------+-------------+------+--------+--------+
| Student_ID | Name        | Age  | Course | City   |
+------------+-------------+------+--------+--------+
|        105 | Karan Mehta |   23 | B.Tech | Meerut |
|        106 | Riya Joshi  |   20 | BCA    | Noida  |
+------------+-------------+------+--------+--------+
2 rows in set
```

Q8. Demonstrate NOT with a condition.

SQL

```
SELECT * FROM Student WHERE NOT City = 'Delhi';
```

Output

```
+------------+--------------+------+--------+---------+
| Student_ID | Name         | Age  | Course | City    |
+------------+--------------+------+--------+---------+
|        101 | Rahul Sharma |   20 | B.Tech | Meerut  |
|        103 | Amit Verma   |   22 | MCA    | Roorkee |
|        105 | Karan Mehta  |   23 | B.Tech | Meerut  |
|        106 | Riya Joshi   |   20 | BCA    | Noida   |
+------------+--------------+------+--------+---------+
4 rows in set
```

Q9. Find students belonging to a specific course AND city.

SQL

```
SELECT * FROM Student WHERE Course = 'B.Tech' AND City = 'Meerut';
```

Output

```
+------------+--------------+------+--------+--------+
| Student_ID | Name         | Age  | Course | City   |
+------------+--------------+------+--------+--------+
|        101 | Rahul Sharma |   20 | B.Tech | Meerut |
|        105 | Karan Mehta  |   23 | B.Tech | Meerut |
+------------+--------------+------+--------+--------+
2 rows in set
```

Q10. Find employees belonging to either of two departments.

SQL

```
SELECT * FROM Employee WHERE Department = 'IT' OR Department = 'HR';
```

Output

```
+--------+-------------+------------+----------+---------+
| Emp_ID | Emp_Name    | Department | Salary   | Bonus   |
+--------+-------------+------------+----------+---------+
|      1 | Arun Kapoor | IT         | 55000.00 | 5000.00 |
|      2 | Meena Iyer  | HR         | 42000.00 |    NULL |
|      4 | Divya Menon | IT         | 61000.00 |    NULL |
|      6 | Rekha Nair  | HR         | 45000.00 | 4000.00 |
+--------+-------------+------------+----------+---------+
4 rows in set
```

C. Comparison Operators

Q11. Demonstrate the = operator.

SQL

```
SELECT * FROM Student WHERE Course = 'BCA';
```

Output

```
+------------+-------------+------+--------+-------+
| Student_ID | Name        | Age  | Course | City  |
+------------+-------------+------+--------+-------+
|        102 | Priya Singh |   21 | BCA    | Delhi |
|        106 | Riya Joshi  |   20 | BCA    | Noida |
|        107 | Rohit Jain  |   22 | BCA    | Delhi |
+------------+-------------+------+--------+-------+
3 rows in set
```

Q12. Demonstrate the > operator.

SQL

```
SELECT * FROM Student WHERE Age > 21;
```

Output

```
+------------+-------------+------+--------+---------+
| Student_ID | Name        | Age  | Course | City    |
+------------+-------------+------+--------+---------+
|        103 | Amit Verma  |   22 | MCA    | Roorkee |
|        105 | Karan Mehta |   23 | B.Tech | Meerut  |
|        107 | Rohit Jain  |   22 | BCA    | Delhi   |
+------------+-------------+------+--------+---------+
3 rows in set
```

Q13. Demonstrate the < operator.

SQL

```
SELECT * FROM Student WHERE Age < 21;
```

Output

```
+------------+--------------+------+--------+--------+
| Student_ID | Name         | Age  | Course | City   |
+------------+--------------+------+--------+--------+
|        101 | Rahul Sharma |   20 | B.Tech | Meerut |
|        104 | Neha Gupta   |   19 | B.Tech | Delhi  |
|        106 | Riya Joshi   |   20 | BCA    | Noida  |
+------------+--------------+------+--------+--------+
3 rows in set
```

Q14. Demonstrate >= and <=.

SQL

```
SELECT Emp_Name, Salary FROM Employee WHERE Salary >= 48000;
SELECT Emp_Name, Salary FROM Employee WHERE Salary <= 42000;
```

Output

```
+--------------+----------+
| Emp_Name     | Salary   |
+--------------+----------+
| Arun Kapoor  | 55000.00 |
| Suresh Patel | 48000.00 |
| Divya Menon  | 61000.00 |
+--------------+----------+
3 rows in set

+------------+----------+
| Emp_Name   | Salary   |
+------------+----------+
| Meena Iyer | 42000.00 |
| Ravi Das   | 30000.00 |
+------------+----------+
2 rows in set
```

Q15. Demonstrate the <> / != operator.

SQL

```
SELECT Name, City FROM Student WHERE City <> 'Delhi';
SELECT Name, City FROM Student WHERE City != 'Meerut';
```

Output

```
+--------------+---------+
| Name         | City    |
+--------------+---------+
| Rahul Sharma | Meerut  |
| Amit Verma   | Roorkee |
| Karan Mehta  | Meerut  |
| Riya Joshi   | Noida   |
+--------------+---------+
4 rows in set

+-------------+---------+
| Name        | City    |
+-------------+---------+
| Priya Singh | Delhi   |
| Amit Verma  | Roorkee |
| Neha Gupta  | Delhi   |
| Riya Joshi  | Noida   |
| Rohit Jain  | Delhi   |
+-------------+---------+
5 rows in set
```

D. Special Operators

Q16. Demonstrate BETWEEN for salary range.

SQL

```
SELECT Emp_Name, Salary FROM Employee WHERE Salary BETWEEN 40000 AND 55000;
```

Output

```
+--------------+----------+
| Emp_Name     | Salary   |
+--------------+----------+
| Arun Kapoor  | 55000.00 |
| Meena Iyer   | 42000.00 |
| Suresh Patel | 48000.00 |
| Rekha Nair   | 45000.00 |
+--------------+----------+
4 rows in set
```

Q17. Demonstrate IN for selecting multiple cities.

SQL

```
SELECT Name, City FROM Student WHERE City IN ('Delhi', 'Meerut', 'Noida');
```

Output

```
+--------------+--------+
| Name         | City   |
+--------------+--------+
| Rahul Sharma | Meerut |
| Priya Singh  | Delhi  |
| Neha Gupta   | Delhi  |
| Karan Mehta  | Meerut |
| Riya Joshi   | Noida  |
| Rohit Jain   | Delhi  |
+--------------+--------+
6 rows in set
```

Q18. Demonstrate NOT IN.

SQL

```
SELECT Name, City FROM Student WHERE City NOT IN ('Delhi', 'Meerut');
```

Output

```
+------------+---------+
| Name       | City    |
+------------+---------+
| Amit Verma | Roorkee |
| Riya Joshi | Noida   |
+------------+---------+
2 rows in set
```

Q19. Demonstrate LIKE for names beginning with a particular letter.

SQL

```
SELECT Name FROM Student WHERE Name LIKE 'R%';      -- starts with R
SELECT Name FROM Student WHERE Name LIKE '%a';      -- ends with a
```

Output

```
+--------------+
| Name         |
+--------------+
| Rahul Sharma |
| Riya Joshi   |
| Rohit Jain   |
+--------------+
3 rows in set

+--------------+
| Name         |
+--------------+
| Rahul Sharma |
| Amit Verma   |
| Neha Gupta   |
| Karan Mehta  |
+--------------+
4 rows in set
```

Q20. Demonstrate IS NULL and IS NOT NULL.

SQL

```
SELECT Emp_Name, Bonus FROM Employee WHERE Bonus IS NULL;
SELECT Emp_Name, Bonus FROM Employee WHERE Bonus IS NOT NULL;
```

Output

```
+-------------+-------+
| Emp_Name    | Bonus |
+-------------+-------+
| Meena Iyer  |  NULL |
| Divya Menon |  NULL |
+-------------+-------+
2 rows in set

+--------------+---------+
| Emp_Name     | Bonus   |
+--------------+---------+
| Arun Kapoor  | 5000.00 |
| Suresh Patel | 3000.00 |
| Ravi Das     | 2000.00 |
| Rekha Nair   | 4000.00 |
+--------------+---------+
4 rows in set
```

E. Set Operations

Q21. Demonstrate UNION using two compatible queries.

SQL

```
-- all distinct cities of students and customers
SELECT City FROM Student
UNION
SELECT City FROM Customer;
```

Output

```
+---------+
| City    |
+---------+
| Meerut  |
| Delhi   |
| Roorkee |
| Noida   |
| Mumbai  |
| Pune    |
+---------+
6 rows in set
```

Q22. Demonstrate UNION ALL.

SQL

```
-- keeps duplicates
SELECT City FROM Student
UNION ALL
SELECT City FROM Customer;
```

Output

```
+---------+
| City    |
+---------+
| Meerut  |
| Delhi   |
| Roorkee |
| Delhi   |
| Meerut  |
| Noida   |
| Delhi   |
| Delhi   |
| Mumbai  |
| Meerut  |
| Pune    |
+---------+
11 rows in set
```

Q23. Demonstrate INTERSECT where supported by the DBMS.

SQL

```
-- cities that have both students and customers
-- (MySQL 8.0.31+ / MariaDB 10.3+)
SELECT City FROM Student
INTERSECT
SELECT City FROM Customer;
```

Output

```
+--------+
| City   |
+--------+
| Meerut |
| Delhi  |
+--------+
2 rows in set
```

Q24. Demonstrate EXCEPT/MINUS where supported.

SQL

```
-- cities that have students but no customers
SELECT City FROM Student
EXCEPT
SELECT City FROM Customer;
```

Output

```
+---------+
| City    |
+---------+
| Roorkee |
| Noida   |
+---------+
2 rows in set
```

Q25. Problem using Arithmetic + Logical + Comparison + Special operators together.

SQL

```
-- Employees from IT/HR/Finance (IN), whose total pay (salary + bonus, arithmetic)
-- is at least 45000 (comparison), who do not have a name starting with D
-- (NOT LIKE), and who have a bonus (IS NOT NULL) in the 40000-70000 salary band (BETWEEN).
SELECT Emp_Name, Department, Salary, Bonus,
       Salary + Bonus AS Total_Pay
FROM Employee
WHERE Department IN ('IT', 'HR', 'Finance')
  AND Salary BETWEEN 40000 AND 70000
  AND Bonus IS NOT NULL
  AND Emp_Name NOT LIKE 'D%'
  AND (Salary + Bonus) >= 45000
ORDER BY Total_Pay DESC;
```

Output

```
+--------------+------------+----------+---------+-----------+
| Emp_Name     | Department | Salary   | Bonus   | Total_Pay |
+--------------+------------+----------+---------+-----------+
| Arun Kapoor  | IT         | 55000.00 | 5000.00 |  60000.00 |
| Suresh Patel | Finance    | 48000.00 | 3000.00 |  51000.00 |
| Rekha Nair   | HR         | 45000.00 | 4000.00 |  49000.00 |
+--------------+------------+----------+---------+-----------+
3 rows in set
```